

A380X SURV controls, STATUS & SWITCHING and pedestal SURV panel

The MFD SURV pages and the pedestal buttons made functional

Branch	feature/a380x-surv-panel
Aircraft	A380X
Commits	3 commits, tip a1c0afd4b (PR 5)
Base	upstream master 2baa2b35e
Status	In develop. PR draft 05.

Specification

- SURV CONTROLS WXR block (WXR, PRED WS, TURB, GAIN, MODE, WX ON VD) writes LVars the radar reads; inhibited only when the selected WXR system is failed.
- STATUS & SWITCHING: WXR DISPLAY shows the real failures; XPDR/TCAS SYS 1 / 2 select
L:A32NX_TRANSPONDER_SYSTEM.
- Pedestal SURV panel: INDICATOR_POWERED was never defined, so no button worked; now DC 1. WXR TAWS SYS 1/2, XPDR TCAS SYS 1/2, G/S MODE, TCAS ABV / BLW / TA ONLY in step with the MFD pages.
- VD messages: NO TERR DATA / NO WX DATA / both, per case.

LVar	Meaning
L:A380X_WXR_OFF , _TURB_OFF , _MODE_MAP , _VD_OFF , _PRED_WS_OFF , _GAIN_MAN	SURV CONTROLS WXR buttons
L:A380X_TCAS_ALERT_LEVEL , _ALT_SELECT (+ _REQUEST)	TCAS settings mirrored for the pedestal

PR description (from the PR draft)

From `pr-drafts/05-a380x-surv.md`, section "Summary of Changes" onwards.

Summary of Changes

MFD SURV CONTROLS: the WXR block was a stub (every button greyed out, "FIXME replace with appropriate events"). Its buttons (WXR, PRED WS, TURB, GAIN, MODE, WX ON VD) now write LVars that the native weather radar reads, and are only inhibited when the selected WXR system is really failed.

STATUS & SWITCHING: the WXR DISPLAY items show the real failure of each system, and the XPDR/TCAS SYS 1 / SYS 2 buttons select the transponder system (`L:A32NX TRANSPONDER_SYSTEM`, the one the TCAS reads) and light from it.

Pedestal SURV panel: the buttons use the backlight indicator template, which runs the button code and lights the button only if `INDICATOR_POWERED` is defined, and `pedestal.xml` never defined it (the EFIS, FCU and KCCU files do), so none of them ever worked. It is now the DC 1 bus, like the backlight. The buttons are linked to the MFD pages: WXR TAWS SYS 1/2 toggle `L:A32NX_WXR_TAWS_SYS_SELECTED`, XPDR TCAS SYS 1/2 set the transponder system, G/S MODE toggles the GPWS G/S inhibit, each lit from the same LVar. TCAS ABV / BLW / TA ONLY follow and change the SURV CONTROLS TCAS settings: the TCAS computer mirrors the settings to LVars for the lights and applies the button requests to the same events the page publishes.

VD messages: TERR SYS OFF removed both terrain and weather from the message logic, it always showed "NO TERR AND WX DATA AVAILABLE". TERR SYS OFF now only gives NO TERR DATA AVAILABLE, a failed or OFF WXR or WX ON VD OFF only gives NO WX DATA AVAILABLE, and the combined message is kept for both.

Known limit: PRED WS and GAIN MAN are written but nothing reads them (the MSFS radar has no predictive windshear or gain).

Cockpit API Changes

New LVars, already documented in `fbw-a380x/docs/a380-simvars.md` (new "Surveillance ATA 34" section, in the commits of this PR):

Var	Type	Description
<code>L:A380X_WXR_OFF</code>	Bool	SURV CONTROLS WXR button: weather radar off
<code>L:A380X_WXR_TURB_OFF</code>	Bool	SURV CONTROLS TURB button: turbulence detection off
<code>L:A380X_WXR_MODE_MAP</code>	Bool	SURV CONTROLS MODE button: MAP mode
<code>L:A380X_WXR_PRED_WS_OFF</code>	Bool	SURV CONTROLS PRED WS button: predictive windshear off (not read by anything yet)
<code>L:A380X_WXR_GAIN_MAN</code>	Bool	SURV CONTROLS GAIN button: manual gain (not read by anything yet)
<code>L:A380X_WXR_VD_OFF</code>	Bool	SURV CONTROLS WX ON VD button: weather on the VD off
<code>L:A380X_TCAS_ALERT_LEVEL</code>	Number	Current TCAS alert level setting of the SURV CONTROLS page (mirrored for the pedestal lights)
<code>L:A380X_TCAS_ALT_SELECT</code>	Number	Current TCAS ABV/BLW altitude selection (mirrored for the pedestal lights)
<code>L:A380X_TCAS_ALERT_LEVEL_REQUEST</code>	Number	Pedestal button request: wanted alert level + 1, 0 = no request (cleared by the TCAS computer)
<code>L:A380X_TCAS_ALT_SELECT_REQUEST</code>	Number	Pedestal button request: wanted altitude selection + 1, 0 = no request

Enum meanings as documented: `L:A380X_TCAS_ALERT_LEVEL` 0 STBY / 1 TA ONLY / 2 TA/RA, `L:A380X_TCAS_ALT_SELECT` 0 NORM / 1 ABV / 2 BLW; the WXR LVars are true for OFF / MANUAL / MAP.

Screenshots (if necessary)

TO ADD: the SURV CONTROLS page (WXR block enabled) and STATUS & SWITCHING before/after; the pedestal SURV panel lit (before: all buttons dark); the VD messages for each case.

References

TO ADD: A380 FCOM/FCTM pages for the SURV panel and the SURV CONTROLS / STATUS & SWITCHING MFD pages (button functions, WXR & TAWS and XPDR & TCAS system selection, TCAS ABV/BLW/TA ONLY, VD no-data messages), with screenshots.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. MFD SURV CONTROLS: WXR, TURB, MODE, WX ON VD buttons change the weather on the ND/VD (with the weather PR); PRED WS / GAIN toggle without effect. With the selected WXR system failed (reset its AESU) the WXR buttons are inhibited.
2. MFD STATUS & SWITCHING: WXR DISPLAY shows the failures; XPDR/TCAS SYS 1 / SYS 2 select the transponder system and light.
3. Pedestal SURV panel: with DC 1 powered the buttons light; WXR TAWS SYS 1/2, XPDR TCAS SYS 1/2, G/S MODE, TCAS ABV / BLW / TA ONLY work and stay in step with the MFD pages (press on the pedestal, look at the MFD, and the other way round). Remove DC 1: lights off.
4. VD messages: TERR SYS OFF -> NO TERR DATA AVAILABLE; WXR OFF/failed or WX ON VD OFF -> NO WX DATA AVAILABLE; both -> the combined message.